3-D Mesh with 2-D Convection

By Byron Farmar

MENG 491 Final Project

1-D Convection

- Apply 1-D Convection Boundry condition to Problem 2.
- Have to create another part of the mesh with line elements.
- Have to apply the boundry condition to stiffness matrix.

$$K_e = \int_\Omega h N^T N \det(J) \, \mathrm{d}\xi$$

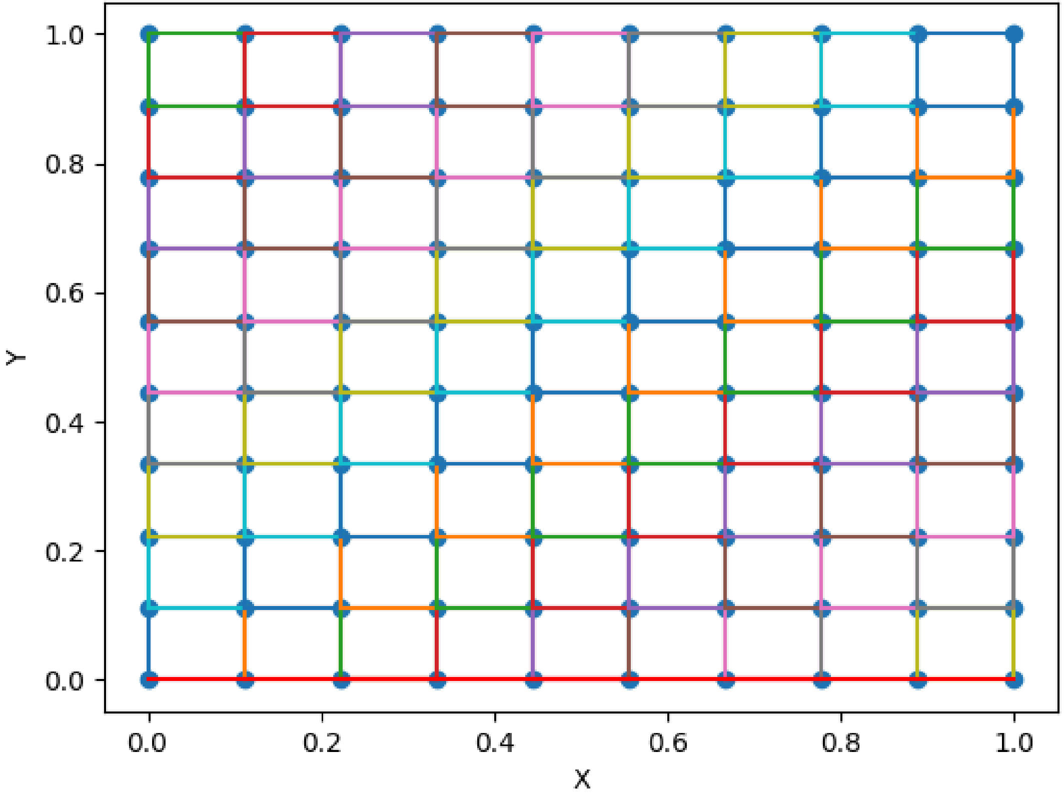

Figure 1: 2-D Mesh with 1-D Convection.

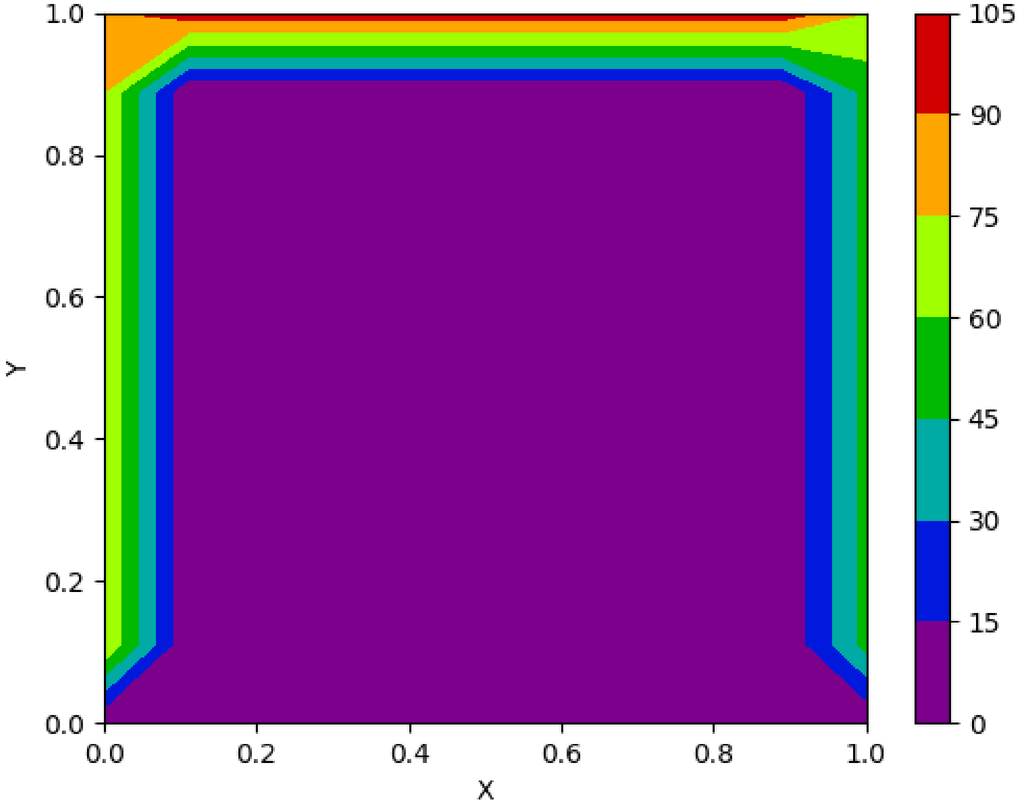

Figure 2: 2-D Boundry Conditions

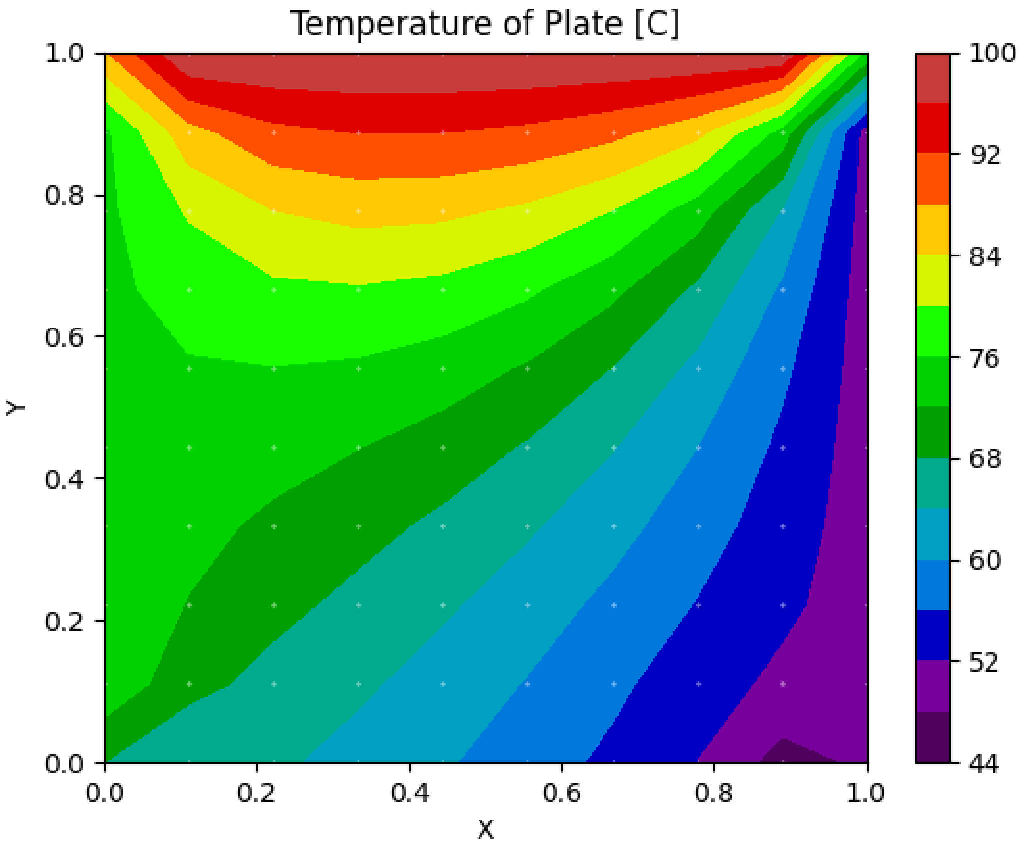

Figure 3: 2-D Results

3-D Mesh

- Must convert K_e to handle 3 dimentions.
- Have to set up mesh correctly.
- Add shape function for hexahedron.

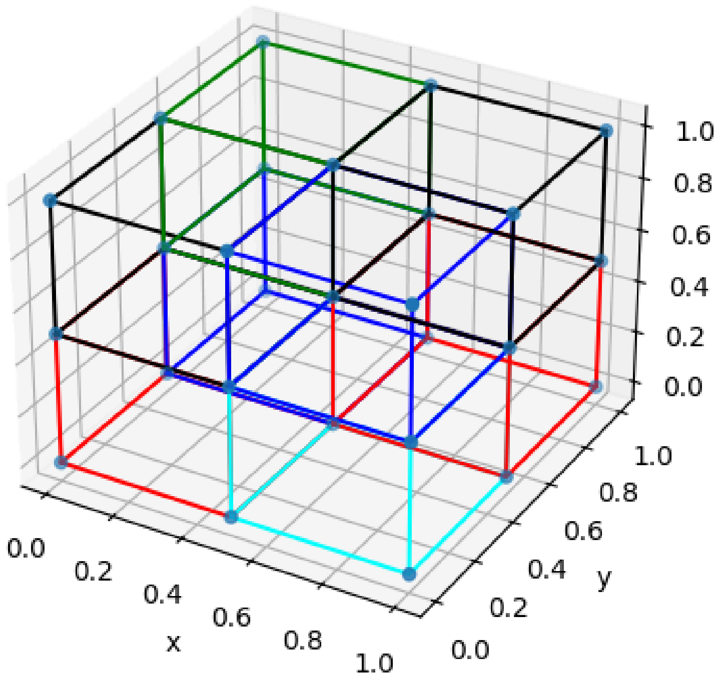

Figure 4: 3-D Mesh

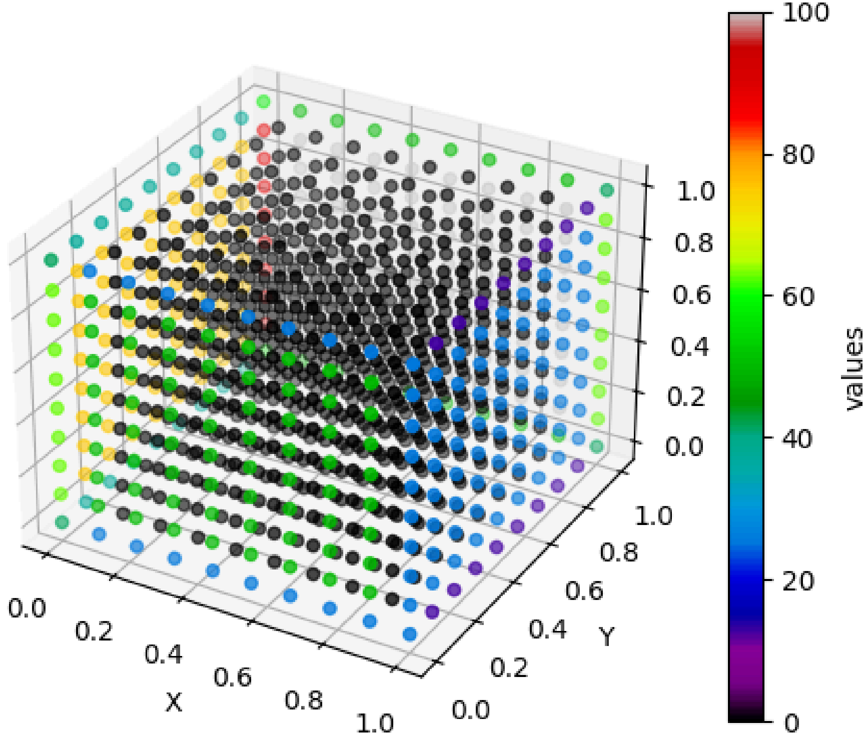

Figure 5: 3-D Boundry Conditions

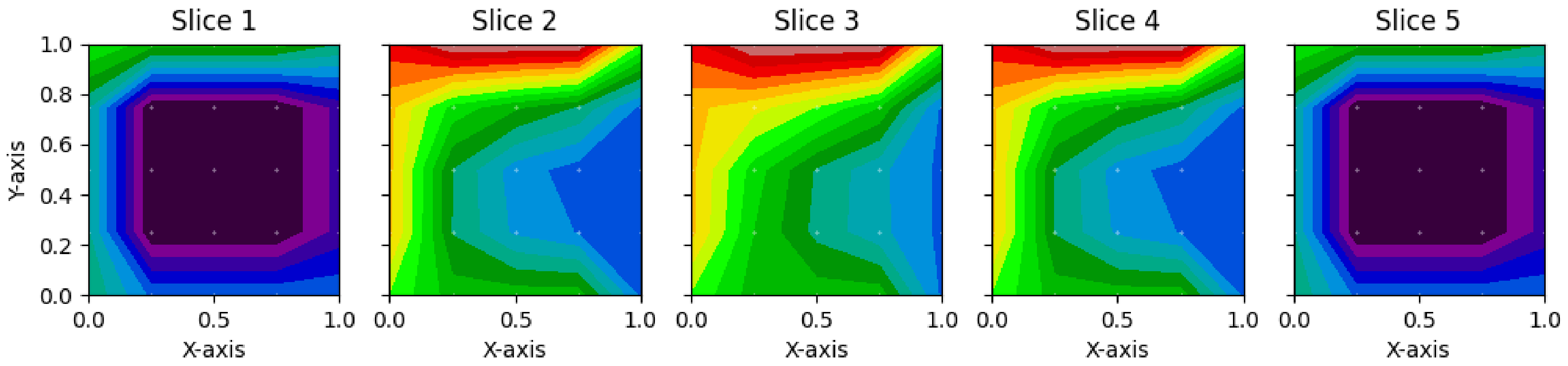

Figure 6: 3-D Results

2-D Convection

- Remove appropriate boundry condition.
- Change convection from line to quadralateral elements.
- Change the modification of K_e to work with quadralateral elements.

$$K_e = \int_\Omega h N^T N \det(J)\, \mathrm{d}\xi\, \mathrm{d}\eta$$

$$\det(J) = \left\| \overrightarrow{r_{,\xi}} \times \overrightarrow{r_{,\eta}} \right\|$$

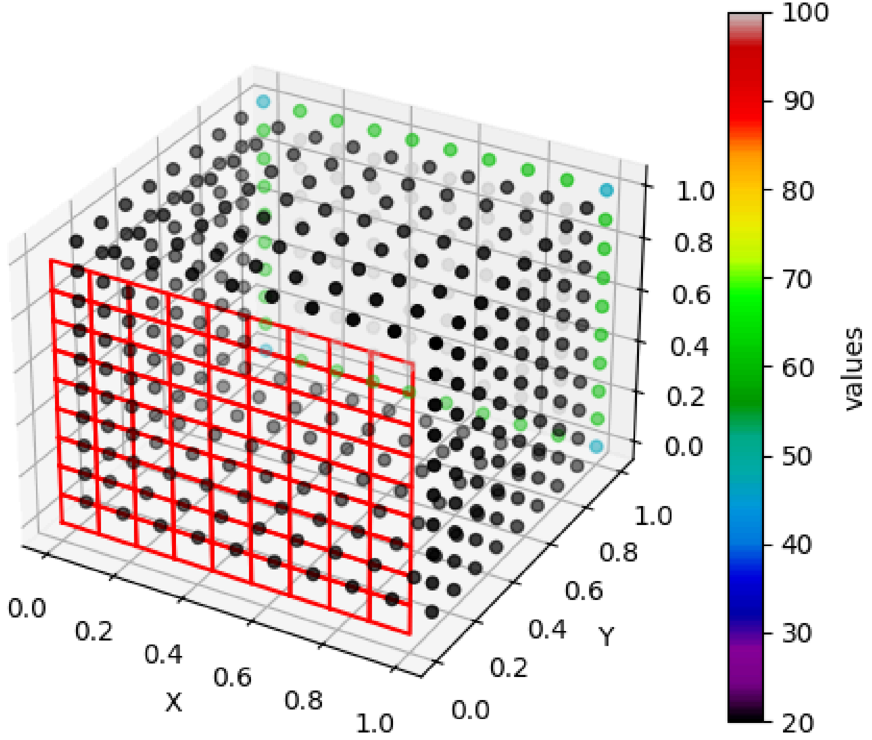

Figure 7: 3-D Convection Boundry Conditions

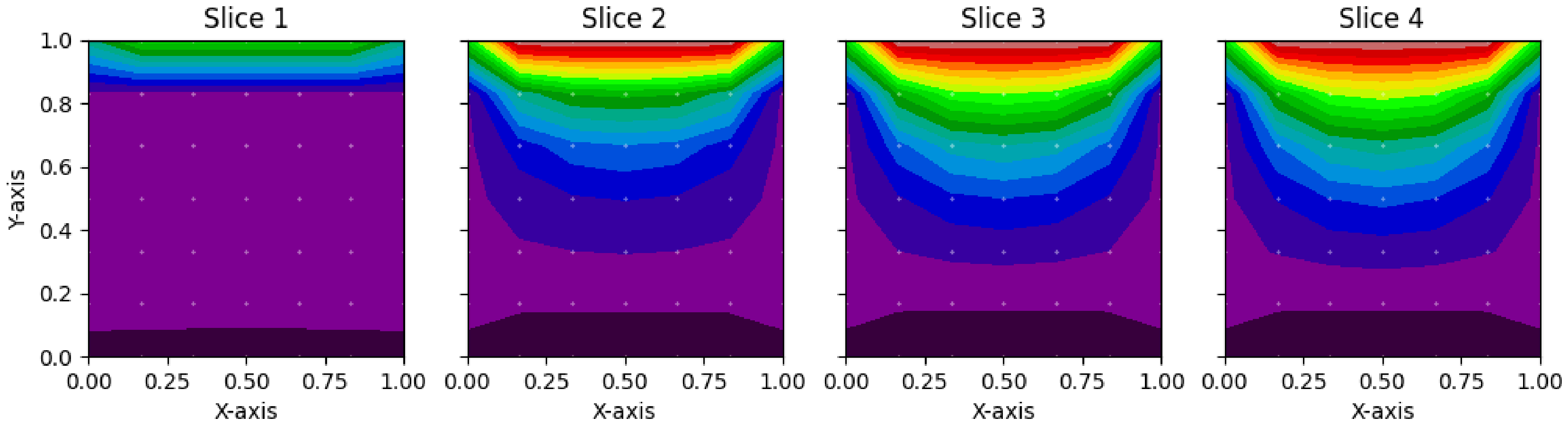

Figure 8: 3-D Convection Results